CHAPTER 1

INTRODUCTION TO EMBEDDED SYSTEMS

Embedded Technology is now in its prime and the wealth of knowledge available is mind blowing. However, most embedded systems engineers have a common complaint. There are no comprehensive resources available over the internet which deal with the various design and implementation issues of this technology. Intellectual property regulations of many corporations are partly to blame for this and also the tendency to keep technical know-how within a restricted group of researchers.

An embedded computer is frequently a computer that is implemented for a particular purpose. In contrast, an average PC computer usually serves a number of purposes: checking email, surfing the internet, listening to music, word processing, etc... However, embedded systems usually only have a single task, or a very small number of related tasks that they are programmed to perform.

Every home has several examples of embedded computers. Any appliance that has a digital clock, for instance, has a small embedded micro-controller that performs no other task than to display the clock. Modern cars have embedded computers onboard that control such things as ignition timing and anti-lock brakes using input from a number of different sensors.

Embedded computers rarely have a generic interface, however. Even if embedded systems have a keypad and an LCD display, they are rarely capable of using many different types of input or output. An example of an embedded system with I/O capability is a security alarm with an LCD status display, and a keypad for entering a password.

An embedded system can be defined as a control system or computer system designed to perform a specific task. Common examples of embedded systems include MP3 players, navigation systems on aircraft and intruder alarm systems. An embedded system can also be defined as a single purpose computer.

Most embedded systems are time critical applications meaning that the embedded system is working in an environment where timing is very important: the results of an operation are only relevant if they take place in a specific time frame. An autopilot in an aircraft is a time critical embedded system. If the autopilot detects that the plane for some reason is going into a stall then it should take steps to correct this within milliseconds or there would be catastrophic results.

1.1 APPLICATIONS OF EMBEDDED SYSTEM

Embedded systems are commonly found in consumer, cooking, industrial, automotive, medical, commercial and military applications.

Telecommunications systems employ numerous embedded systems from telephone switches for the network to cell phones at the end user. Computer networking uses dedicated routers and network bridges to route data.

Consumer electronics include MP3 players, mobile phones, videogame consoles, digital cameras, GPS receivers, and printers. Household appliances, such as microwave ovens, washing machines and dishwashers, include embedded systems to provide flexibility, efficiency and features. Advanced HVAC systems use networked thermostats to more accurately and efficiently control temperature that can change by time of day and season. Home automation uses wired- and wireless-networking that can be used to control lights, climate, security, audio/visual, surveillance, etc., all of which use embedded devices for sensing and controlling.

Transportation systems from flight to automobiles increasingly use embedded systems. New airplanes contain advanced avionics such as inertial guidance systems and GPS receivers that also have considerable safety requirements. Various electric motors brushless DC motors, induction motors and DC motors use electric/electronic motor controllers. Automobiles, electric vehicles, and hybrid vehicles increasingly use embedded systems to maximize efficiency and reduce pollution. Other automotive safety systems include anti-lock braking system (ABS), Electronic Stability Control (ESC/ESP), traction control (TCS) and automatic four-wheel drive.

Medical equipment uses embedded systems for vital signs monitoring, electronic stethoscopes for amplifying sounds, and various medical imaging (PET, SPECT, CT, and MRI) for non-invasive internal inspections. Embedded systems within medical equipment are often powered by industrial computers.[9]

Embedded systems are used in transportation, fire safety, safety and security, medical applications and life critical systems, as these systems can be isolated from hacking and thus, be more reliable. For fire safety, the systems can be designed to have greater ability to handle higher temperatures and continue to operate. In dealing with security, the embedded systems can be self-sufficient and be able to deal with cut electrical and communication systems.

1.2 CHARACTERISTICS OF EMBEDDED SYSTEM

Embedded systems are designed to do some specific task, rather than be a general-purpose computer for multiple tasks. Some also have real-time performance constraints that must be met, for reasons such as safety and usability; others may have low or no performance requirements, allowing the system hardware to be simplified to reduce costs.

Embedded systems are not always standalone devices. Many embedded systems consist of small parts within a larger device that serves a more general purpose. For example, the Gibson Robot Guitar features an embedded system for tuning the strings, but the overall purpose of the Robot Guitar is, of course, to play music. Similarly, an embedded system in an automobile provides a specific function as a subsystem of the car itself.

Embedded systems range from no user interface at all, in systems dedicated only to one task, to complex graphical user interfaces that resemble modern computer desktop operating systems. Simple embedded devices use buttons, LEDs, graphic or character LCDs (HD44780 LCD for example) with a simple menu system.

More sophisticated devices which use a graphical screen with touch sensing or screen-edge buttons provide flexibility while minimizing space used: the meaning of the buttons can change with the screen, and selection involves the natural behavior of pointing at what is desired. Handheld systems often have a screen with a "joystick button" for a pointing device.

Some systems provide user interface remotely with the help of a serial (e.g. RS-232, USB, I²C, etc.) or network (e.g. Ethernet) connection. This approach gives several advantages: extends the capabilities of embedded system, avoids the cost of a display, simplifies BSP and allows one to build a rich user interface on the PC. A good example of this is the combination of an embedded web server running on an embedded device (such as an IP camera) or a network router. The user interface is displayed in a web browser on a PC connected to the device, therefore needing no software to be installed.

1.3 PROCESSORS IN EMBEDDED SYSTEMS

Embedded processors can be broken into two broad categories. Ordinary microprocessors Embedded processors can be broken into two broad categories. Ordinary microprocessors (μP) use separate integrated circuits for memory and peripherals. Microcontrollers (μC) have on-chip peripherals, thus reducing power consumption, size and cost. In contrast to the personal computer market, many different basic CPU architectures are used, since software is custom-developed for an application and is not a commodity product installed by the end user. Both Von Neumann as well as various degrees of Harvard architectures are used. RISC as well as non-RISC processors are found. Word lengths vary from 4-bit to 64-bits and beyond, although the most typical remain 8/16-bit. Most architectures come in a large number of different variants and shapes, many of which are also manufactured by several different companies.

Numerous microcontrollers have been developed for embedded systems use. General-purpose microprocessors are also used in embedded systems, but generally require more support circuitry than microcontrollers.

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1.4 DEBUGGING IN EMBEDDED SYSTEMS

Embedded debugging may be performed at different levels, depending on the facilities available. The different metrics that characterize the different forms of embedded debugging are: does it slow down the main application, how close is the debugged system or application to the actual system or application, how expressive are the triggers that I can set for debugging (e.g., I want to inspect the memory when a particular program counter value is reached), and what can I inspect in the debugging process (such as, only memory, or memory and registers, etc.).

From simplest to most sophisticated they can be roughly grouped into the following areas:

Interactive resident debugging, using the simple shell provided by the embedded operating system (e.g. Forth and Basic)

External debugging using logging or serial port output to trace operation using either a monitor in flash or using a debug server like the Remedy Debugger which even works for heterogeneous multicore systems.

An in-circuit debugger (ICD), a hardware device that connects to the microprocessor via a JTAG or Nexus interface. This allows the operation of the microprocessor to be controlled externally, but is typically restricted to specific debugging capabilities in the processor.

An in-circuit emulator (ICE) replaces the microprocessor with a simulated equivalent, providing full control over all aspects of the microprocessor.

A complete emulator provides a simulation of all aspects of the hardware, allowing all of it to be controlled and modified, and allowing debugging on a normal PC. The downsides are expense and slow operation, in some cases up to 100 times slower than the final system.

For SoC designs, the typical approach is to verify and debug the design on an FPGA prototype board. Tools such as Certus[11] are used to insert probes in the FPGA RTL that make signals available for observation. This is used to debug hardware, firmware and software interactions across multiple FPGA with capabilities similar to a logic analyzer.

Unless restricted to external debugging, the programmer can typically load and run software through the tools, view the code running in the processor, and start or stop its operation. The view of the code may be as HLL source-code, assembly code or mixture of both.

Because an embedded system is often composed of a wide variety of elements, the debugging strategy may vary. For instance, debugging a software- (and microprocessor-) centric embedded system is different from debugging an embedded system where most of the processing is performed by peripherals (DSP, FPGA, and co-processor). An increasing number of embedded systems today use more than one single processor core. A common problem with multi-core development is the proper synchronization of software execution. In such a case, the embedded system design may wish to check the data traffic on the busses between the processor cores, which requires very low-level debugging, at signal/bus level, with a logic analyzer, for instance.

1.5 RELIABILITY

Embedded systems often reside in machines that are expected to run continuously for years without errors, and in some cases recover by themselves if an error occurs. Therefore, the software is usually developed and tested more carefully than that for personal computers, and unreliable mechanical moving parts such as disk drives, switches or buttons are avoided.

Specific reliability issues may include:

- The system cannot safely be shut down for repair, or it is too inaccessible to repair. Examples include space systems, undersea cables, navigational beacons, bore-hole systems, and automobiles.
- The system must be kept running for safety reasons. "Limp modes" are less tolerable. Often backups are selected by an operator. Examples include aircraft navigation, reactor control systems, safety-critical chemical factory controls, train signals.
- The system will lose large amounts of money when shut down: Telephone switches, factory controls, bridge and elevator controls, funds transfer and market making, automated sales and service.

A variety of techniques are used, sometimes in combination, to recover from errors—both software bugs such as memory leaks, and also soft errors in the hardware:

- watchdog timer that resets the computer unless the software periodically notifies the watchdog subsystems with redundant spares that can be switched over to software "limp modes" that provide partial function
- Designing with a Trusted Computing Base (TCB) architecture[12] ensures a highly secure & reliable system environment
- A hypervisor designed for embedded systems, is able to provide secure encapsulation for any subsystem component, so that a compromised software component cannot interfere with other subsystems, or privileged-level system software. This encapsulation keeps faults from propagating from one subsystem to another, improving reliability. This may also allow a subsystem to be automatically shut down and restarted on fault detection.
- Immunity Aware Programming

1.6 TRACING

Real-time operating systems (RTOS) often supports tracing of operating system events. A graphical view is presented by a host PC tool, based on a recording of the system behavior. The trace recording can be performed in software, by the RTOS, or by special tracing hardware. RTOS tracing allows developers to understand timing and performance issues of the software system and gives a good understanding of the high-level system behaviors. Commercial tools like RTXC Quadros or IAR Systems exists.

CHAPTER 2

WIRELESS CHARGING STATION FOR ELECTRIC VEHICLE

INTRODUCTION

In the rapidly evolving landscape of electric vehicles (EVs), the demand for innovative and efficient charging solutions has become increasingly paramount. One such groundbreaking technology that has emerged to address this need is wireless charging stations for electric vehicles. Unlike traditional plug-in charging systems, wireless charging offers a seamless and convenient approach to powering up EVs, marking a significant step towards the future of sustainable transportation. As we delve into the intricacies of wireless charging, it becomes evident that this technology holds the potential to revolutionize the EV charging infrastructure, offering users a more streamlined and user-friendly experience.

SYSTEM DESIGN

Designing a wireless charging station for electric vehicles involves several key components and considerations to ensure efficiency, safety, and compatibility. Below is an overview of the system design for a wireless charging station:

1. **Power Electronics:**

- The heart of the wireless charging system is the power electronics, including the power inverter and rectifier. The power inverter converts AC power from the grid to high-frequency AC, while the rectifier converts it back to DC power for charging the vehicle's battery.

2. **Grid Connection:**

- The charging station needs to be connected to the electrical grid. This connection involves transformers and switchgear to regulate and manage the power flow from the grid to the charging station.

3. **Power Transfer System:**

- Wireless charging relies on inductive or resonant coupling for power transfer. This involves a transmitting coil in the charging station and a receiving coil in the electric vehicle. The power transfer system should be designed for high efficiency and minimal energy losses during the wireless charging process.

4. **Communication Protocols:**

- Implementing communication protocols between the charging station and the electric vehicle is crucial. This includes protocols for authentication, billing, and data exchange. Common standards like ISO/IEC 15118 for communication between electric vehicles and charging stations should be considered.

5. **Safety Systems:**

- Safety is paramount in electric vehicle charging. The system should include various safety features such as overcurrent protection, overvoltage protection, temperature monitoring, and fault detection to ensure safe and reliable operation.

6. **User Interface:**

- A user-friendly interface is essential for a positive user experience. This includes a display for relevant information, user authentication mechanisms, and easy-to-use connectors or pads for parking the vehicle over the charging station.

7. **Integration with Smart Grid:**

- To enhance efficiency and grid management, integrating the charging station with smart grid technologies is advisable. This allows for load balancing, demand response, and optimal utilization of renewable energy sources.

8. **Modular Design:**

- A modular design allows for scalability and easy maintenance. Components such as power modules, communication units, and safety systems should be designed as modular units for easier replacement and upgrades.

9. **Environmental Considerations:**

- Considerations for environmental impact and sustainability should be integrated into the design. This includes materials used, energy efficiency, and end-of-life recycling options.

10. **Compliance with Standards:**

- Ensuring that the wireless charging station complies with industry standards and regulations is crucial. This includes safety standards, electromagnetic compatibility (EMC) standards, and any regional or international regulations.

11. **Security Measures:**

- Implementing robust security measures is essential to protect against unauthorized access, cyber threats, and ensure the integrity of the charging process and user data.

By addressing these key components and considerations, the system design of a wireless charging station for electric vehicles can be optimized for efficiency, safety, and seamless integration into the evolving landscape of electric mobility.

OBJECTIVES

The objectives of a wireless charging station for electric vehicles (EVs) are multifaceted and aim to address various aspects to enhance the overall usability, efficiency, and sustainability of the charging infrastructure. Here are the key objectives:

1. **Convenience and User-Friendliness:**

- Simplify the charging process by eliminating the need for physical connectors and cables, providing users with a more convenient and user-friendly experience.

2. **Efficient Power Transfer:**

   - Optimize the wireless power transfer system to ensure high efficiency, minimizing energy losses during the charging process and maximizing the amount of energy transferred to the electric vehicle.

3. **Standardization and Compatibility:**

   - Establish and adhere to industry standards to ensure compatibility between different electric vehicles and charging stations, promoting interoperability and a seamless user experience.

4. **Safety Assurance:**

   - Prioritize safety by implementing robust safety features, such as overcurrent protection, overvoltage protection, temperature monitoring, and fault detection, to guarantee the safety of users and the integrity of the charging process.

5. **Scalability and Integration:**

   - Design the wireless charging station with scalability in mind, allowing for easy integration into various environments and accommodating future advancements in electric vehicle technology.

6. **Smart Grid Integration:**

   - Facilitate integration with smart grid technologies to enable features like demand response, load balancing, and optimal utilization of renewable energy sources, contributing to overall energy efficiency.

7. **Fast and Efficient Charging:**

- Provide fast and efficient charging capabilities to reduce the time required for replenishing the electric vehicle's battery, making electric vehicles more practical for everyday use.

8. **Environmental Sustainability:**

- Incorporate environmentally friendly materials and energy-efficient components to minimize the environmental impact of the charging station and promote sustainability in electric transportation.

9. **Reliable Communication:**

- Establish reliable communication protocols between the charging station and electric vehicles for tasks such as authentication, billing, and data exchange, ensuring secure and seamless interactions.

10. **Cost-Effectiveness:**

- Strive for cost-effective solutions in both the deployment and maintenance of wireless charging stations, making electric vehicles more economically viable for a broader range of users.

11. **Adaptability to Various Locations:**

- Design charging stations that can adapt to different locations, such as homes, public parking spaces, and commercial areas, accommodating diverse user needs and supporting the widespread adoption of electric vehicles.

12. **User Authentication and Billing:**

- Implement secure user authentication mechanisms and billing systems to ensure that only authorized users can access and use the charging station, supporting fair and transparent payment processes.

13. **Long-Term Reliability:**

- Design the wireless charging station with a focus on long-term reliability, durability, and minimal maintenance requirements, contributing to the overall sustainability of the charging infrastructure.

By addressing these objectives, the development and deployment of wireless charging stations for electric vehicles can significantly contribute to the advancement and widespread adoption of electric mobility while enhancing the overall user experience and environmental sustainability.

WORKING

The working of a wireless charging station for electric vehicles involves several key components and a carefully orchestrated process to transfer power efficiently from the charging station to the vehicle without the need for physical cables. Here's a general overview of the working principles:

1. **Power Generation and Grid Connection:**

- The wireless charging station is connected to the electrical grid to obtain the necessary power. Transformers and other grid connection components regulate the voltage to the required levels for the charging process.

2. **Power Electronics and Inverter:**

- The incoming AC power from the grid is processed by power electronics, including an inverter, which converts AC power to high-frequency AC.

3. **Transmitting Coil in the Charging Station:**

   - The charging station is equipped with a transmitting coil, often made of copper, which is part of the primary side of the inductive coupling system. This coil generates a magnetic field when supplied with high-frequency AC power.

4. **Receiving Coil in the Electric Vehicle:**

   - The electric vehicle is equipped with a receiving coil, usually positioned on the vehicle's undercarriage. This coil is part of the secondary side of the inductive coupling system and is responsible for capturing the magnetic field generated by the transmitting coil.

5. **Magnetic Field Induction:**

   - When the transmitting coil in the charging station generates a magnetic field, it induces a voltage in the receiving coil on the electric vehicle. This process is based on electromagnetic induction principles.

6. **Power Transfer and Rectification:**

   - The induced voltage in the receiving coil is then rectified to convert it back to DC power. This DC power is used to charge the electric vehicle's battery.

7. **Communication and Control Systems:**

   - Communication protocols are employed between the charging station and the electric vehicle for tasks such as authentication, billing, and monitoring. These protocols ensure secure and reliable interactions.

8. **Safety Features:**

   - The wireless charging system includes safety features such as temperature monitoring, overcurrent protection, and overvoltage protection to ensure safe and reliable charging.

9. **User Interface:**

- A user interface is provided on the charging station, which may include a display screen for relevant information, user authentication mechanisms, and indicators to guide the driver during the charging process.

10. **Automated Charging Process:**

- Once the electric vehicle is properly aligned with the charging station, the charging process begins automatically. The driver doesn't need to handle any physical connectors, making the process seamless and user-friendly.

11. **End of Charging and Disconnect:**

- The charging station monitors the charging progress and automatically stops the charging process when the electric vehicle's battery is fully charged or reaches the desired charging level. The driver can then disconnect and drive away.

The wireless charging station's working mechanism is designed to be efficient, safe, and user-friendly, contributing to the growing adoption of electric vehicles by providing a convenient and streamlined charging experience.

ADVANTAGES

Wireless charging stations for electric vehicles offer several advantages, contributing to the convenience, efficiency, and widespread adoption of electric mobility. Here are some key advantages:

1. **Convenience and User-Friendly Experience:**

- Eliminates the need for physical cables and connectors, providing a hassle-free and more convenient charging experience. Users can simply park over the charging station, and the charging process initiates automatically.

2. **Reduced Wear and Tear:**

- Wireless charging eliminates the wear and tear associated with traditional plug-in charging systems. There are no physical connectors that can degrade over time, leading to increased reliability and longevity of the charging infrastructure.

3. **Enhanced Safety:**

- Minimizes safety risks associated with physical connectors, such as tripping hazards or exposure to live electrical contacts. Wireless charging stations also incorporate safety features like overcurrent protection and temperature monitoring to ensure secure charging.

4. **Streamlined Integration into Urban Environments:**

- Wireless charging stations can be seamlessly integrated into various urban environments, including residential areas, public parking lots, and commercial spaces. This flexibility supports the widespread deployment of electric vehicle infrastructure.

5. **Scalability and Adaptability:**

- Designed for scalability, wireless charging infrastructure can be easily adapted to different locations and environments. This adaptability promotes the growth of electric vehicle charging networks.

6. **Reduced Energy Losses:**

- Wireless charging technologies are continually improving in efficiency, leading to reduced energy losses during the power transfer process. This efficiency contributes to overall energy savings and a more sustainable charging infrastructure.

7. **Smart Grid Integration:**

- Wireless charging stations can be integrated with smart grid technologies, allowing for features like demand response, load balancing, and optimization of energy usage. This integration enhances the overall efficiency of the electric grid.

8. **Flexibility in Parking Alignment:**

- Wireless charging systems often offer flexibility in parking alignment, allowing for slight misalignments between the vehicle and the charging pad. This flexibility enhances the ease of use and ensures a successful charging connection.

9. **Enhanced Aesthetics and Urban Planning:**

- Without the need for visible charging cables and infrastructure, wireless charging stations contribute to cleaner and more aesthetically pleasing urban environments. This can encourage municipalities and businesses to integrate electric vehicle infrastructure into their planning seamlessly.

10. **Positive Environmental Impact:**

- By promoting the adoption of electric vehicles and reducing the barriers to charging, wireless charging stations contribute to the overall reduction of greenhouse gas emissions and dependence on fossil fuels, supporting environmental sustainability.

11. **Improved Resilience and Reliability:**

- The absence of physical connectors and the associated wear and tear enhances the reliability and resilience of wireless charging stations. This makes them suitable for heavy usage and minimizes maintenance requirements.

12. **Enhanced Market Appeal for Electric Vehicles:**

- The convenience and ease of use associated with wireless charging stations can enhance the market appeal of electric vehicles, attracting a broader range of consumers to embrace electric mobility.

As technology advances and becomes more widespread, the advantages of wireless charging stations for electric vehicles will likely continue to grow, contributing to the ongoing shift towards sustainable transportation.

APPLICATIONS

Wireless charging stations for electric vehicles have a variety of applications across different sectors and environments. Here are some key applications:

1. **Residential Use:**

   - Homeowners can install wireless charging stations in their garages or driveways, providing a convenient and hassle-free way to charge electric vehicles overnight. This is particularly beneficial for those who may not have easy access to traditional charging infrastructure.

2. **Public Parking Spaces:**

   - Wireless charging stations can be installed in public parking lots and spaces, allowing electric vehicle users to charge their vehicles while shopping, working, or running errands. This enhances the accessibility of charging infrastructure for users without home charging capabilities.

3. **Commercial and Workplace Charging:**

   - Businesses can integrate wireless charging stations into their parking facilities, offering employees and visitors a convenient way to charge their electric vehicles while at work. This promotes sustainable transportation options for commuters.

4. **Fleet Charging:**

- Wireless charging is suitable for electric vehicle fleets, such as those used by delivery services or taxi companies. Charging stations at centralized locations can efficiently charge multiple vehicles throughout the day without the need for manual connections.

5. **Public Transportation Hubs:**

- Bus stations, train stations, and other public transportation hubs can incorporate wireless charging infrastructure to support electric buses and other electric public transport vehicles. This helps reduce emissions in urban areas and provides a reliable charging solution for transit fleets.

6. **Retail and Shopping Centers:**

- Installing wireless charging stations in retail and shopping center parking lots encourages electric vehicle users to visit these locations, providing an additional amenity for customers. This also aligns with corporate sustainability initiatives.

7. **Hotels and Resorts:**

- Hotels and resorts can offer wireless charging facilities to guests with electric vehicles. This enhances the appeal of the establishment for environmentally conscious travelers and provides an added service.

8. **Tourist Attractions and Recreation Areas:**

- Popular tourist attractions, national parks, and recreation areas can benefit from wireless charging stations to accommodate electric vehicle users exploring these destinations. This supports sustainable tourism and minimizes environmental impact.

9. **Smart Cities and Urban Planning:**

- Wireless charging stations play a role in smart city initiatives, contributing to the development of sustainable and connected urban environments. Integrating these stations into urban planning helps create cleaner and more efficient transportation systems.

10. **Emergency Services and Public Fleets:**

- Police departments, fire stations, and other emergency services can utilize wireless charging for their electric vehicle fleets. This ensures rapid deployment without the need for manual charging connections during critical situations.

11. **Highways and Expressways:**

- Installing wireless charging infrastructure along highways and expressways allows for dynamic charging on the go. This can alleviate range anxiety for electric vehicle users during long-distance travel.

12. **Rural and Remote Areas:**

- Wireless charging can be particularly beneficial in rural and remote areas where establishing traditional charging infrastructure may be challenging. It provides an alternative for electric vehicle users in less densely populated regions.

The versatility of wireless charging stations makes them applicable in various contexts, contributing to the widespread adoption and integration of electric vehicles into different facets of society and industry.

FUTURE SCOPE

The future scope of wireless charging stations for electric vehicles is promising, with ongoing advancements in technology and a growing emphasis on sustainable transportation. Several trends and developments suggest a bright future for the deployment and evolution of wireless charging infrastructure:

1. **Increased Adoption and Market Growth:**

- As electric vehicle adoption continues to rise, the demand for convenient and user-friendly charging solutions is expected to drive the growth of wireless charging stations. The technology's appeal lies in its ease of use and potential to further simplify the charging process for EV users.

2. **Technological Advancements:**

- Ongoing research and development efforts are likely to result in technological advancements, leading to higher efficiency, faster charging rates, and improved interoperability between different electric vehicle models and charging stations.

3. **Standardization Efforts:**

- Standardization of wireless charging protocols is crucial for interoperability and mass adoption. Industry efforts to establish and adhere to common standards will contribute to the seamless integration of wireless charging infrastructure into various environments.

4. **Integration with Smart Grids:**

- The integration of wireless charging stations with smart grids will enhance energy management, allowing for better load balancing, demand response, and optimization of renewable energy sources. This integration aligns with the broader goal of creating intelligent and sustainable energy ecosystems.

5. **Dynamic Charging on the Move:**

- Advancements in dynamic wireless charging technologies may enable charging while the electric vehicle is in motion, particularly on highways or dedicated lanes. This could significantly reduce range anxiety and enhance the practicality of electric vehicles for long-distance travel.

6. **Commercial Fleets and Autonomous Vehicles:**

- Wireless charging is poised to play a vital role in supporting commercial fleets of electric vehicles, such as delivery trucks and autonomous vehicles. Seamless charging can optimize the operation of these fleets, leading to increased efficiency and reduced downtime.

7. **Incorporation into Urban Infrastructure:**

- Future urban planning initiatives are likely to integrate wireless charging infrastructure into the design of smart cities, making it a seamless part of parking spaces, public transportation hubs, and commercial areas.

8. **Enhanced Energy Storage Integration:**

- Wireless charging stations may be combined with advanced energy storage solutions, such as stationary batteries or supercapacitors, to store energy during periods of low demand and release it during peak charging times, further optimizing energy usage.

9. **Increased Range for Electric Vehicles:**

- The development of higher-capacity batteries and improved energy density may contribute to increased electric vehicle ranges. Wireless charging stations will complement this advancement by providing efficient and convenient charging options.

10. **Global Infrastructure Expansion:**

- Continued investment in electric vehicle charging infrastructure, including wireless charging stations, is expected on a global scale. Governments, businesses, and organizations are likely to collaborate to expand the charging network, supporting the transition to electric mobility.

11. **Environmental and Regulatory Support:**

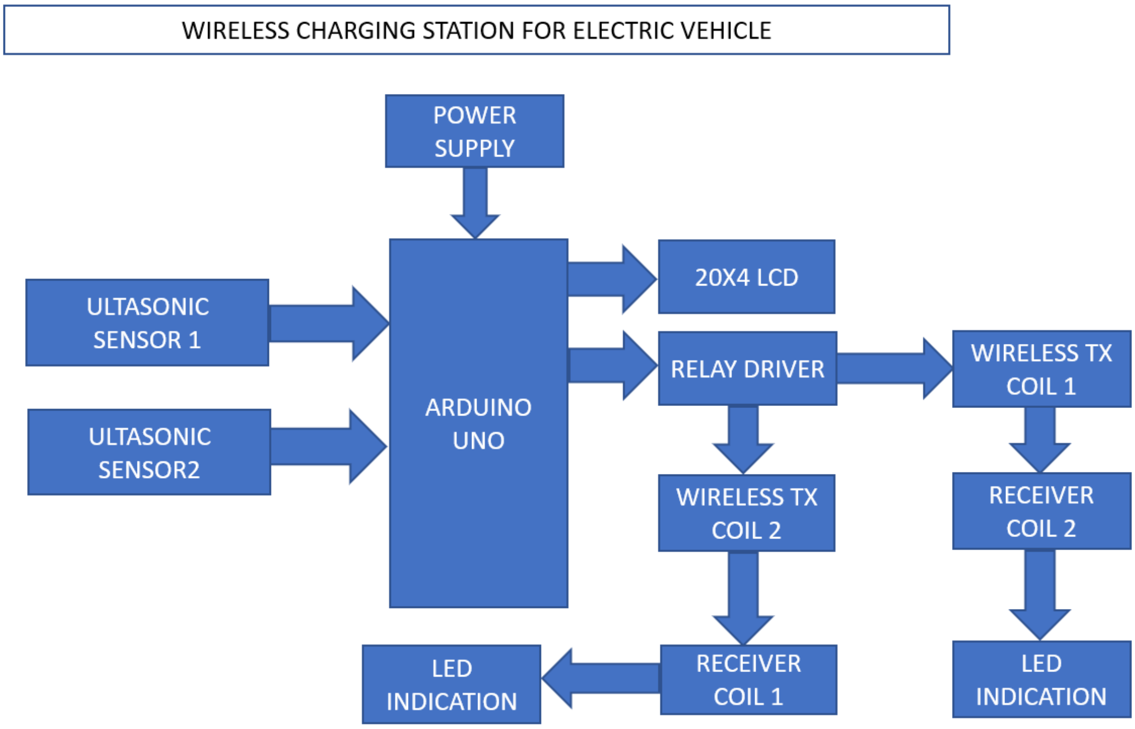
- Growing environmental awareness and regulatory initiatives promoting clean energy and sustainable transportation will further support the expansion of wireless charging infrastructure, creating a conducive environment for investment and development.

12. **Integration with Renewable Energy Sources:**

- The integration of wireless charging stations with renewable energy sources, such as solar and wind power, will enhance the sustainability of electric vehicle charging. This synergy aligns with efforts to reduce the carbon footprint of transportation.

The future scope of wireless charging stations for electric vehicles is dynamic, with ongoing innovation and a collective effort towards building a robust, sustainable, and widely accessible charging infrastructure. As technology continues to evolve, the vision of seamless and efficient electric vehicle charging is likely to become an integral part of the future transportation landscape.

BLOCK DIAGRAM



CHAPTER 3

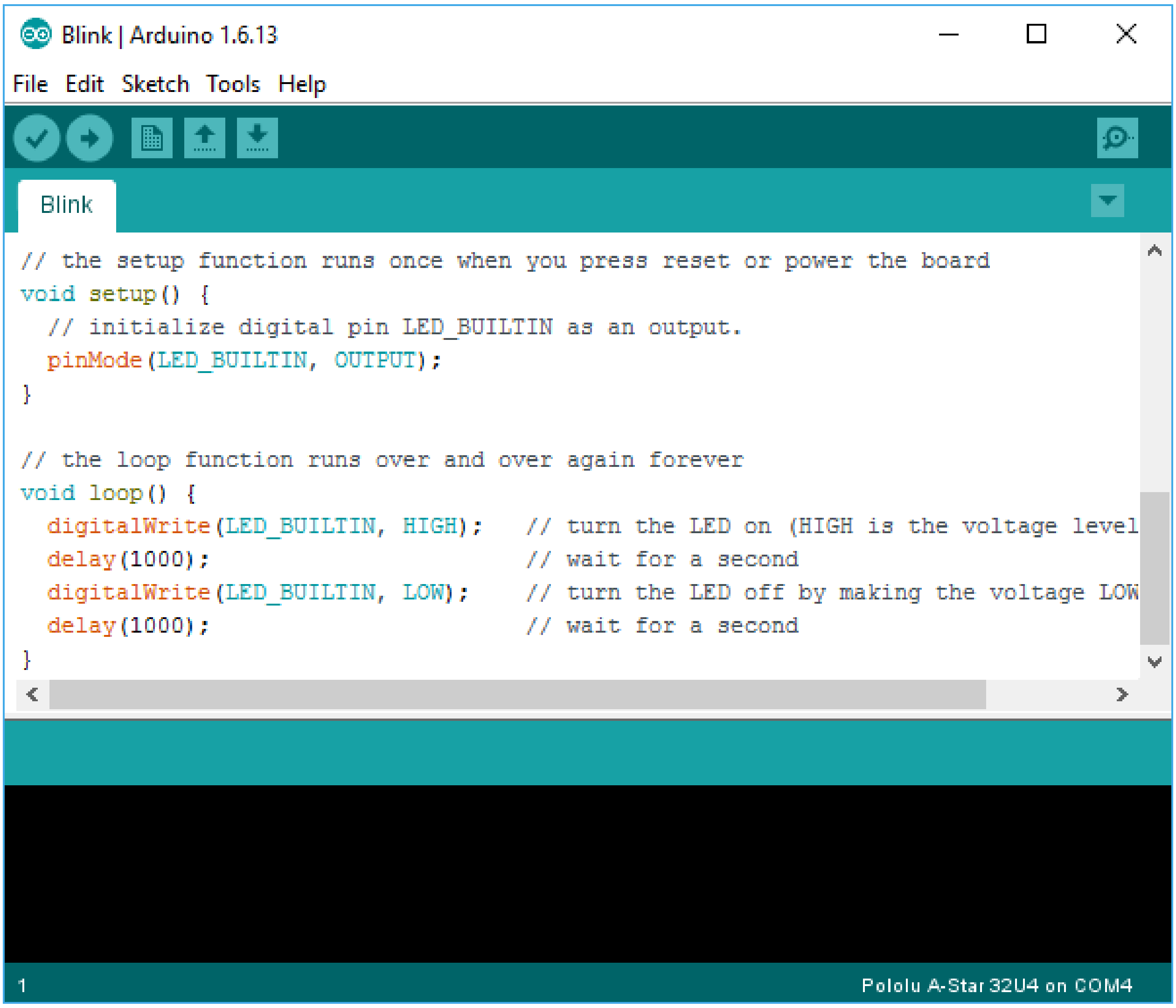
SOFTWARE REQUIREMENTS

Software used in this project for uploading code onto Arduino is Arduino IDE.

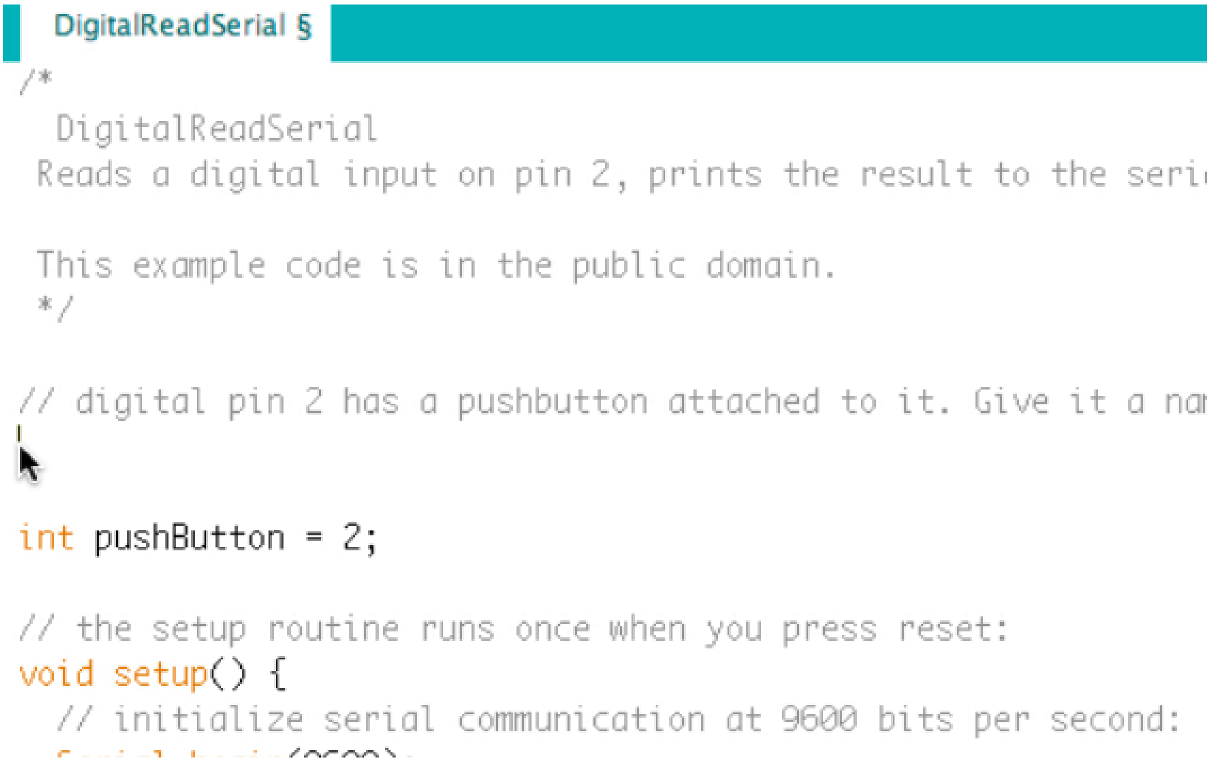
3.1 INTRODUCTION TO ARDUINO IDE

IDE stands for Integrated Development Environment. Pretty fancy sounding, and should make you feel smart any time you use it. The IDE is a text editor-like program that allows you to write Arduino code. When you open the Arduino program, you are opening the IDE. It is intentionally streamlined to keep things as simple and straightforward as possible. When you save a file in Arduino, the file is called a sketch – a sketch is where you save the computer code you have written. The coding language that Arduino uses is very much like C++ ("see plus plus"), which is a common language in the world of computing. The code you learn to write for Arduino will be very similar to the code you write in any other computer language – all the basic concepts remain the same – it is just a matter of learning a new dialect should you pursue other programming languages.

The code you write is "human readable", that is, it will make sense to you (sometimes), and will be organized for a human to follow. Part of the job of the IDE is to take the human readable code and translate it into machine-readable code to be executed by the Arduino. This process is called compiling. The process of compiling is seamless to the user. All you have to do is press a button. If you have errors in your computer code, the compiler will display an error message at the bottom of the IDE and highlight the line of code that seems to be the issue. The error message is meant to help you identify what you might have done wrong – sometimes the message is very explicit, like saying, “Hey – you forget a semicolon”, sometimes the error message is vague. Why be concerned with a semicolon you ask? A semicolon is part of the Arduino language syntax, the rules that govern how the code is written. It is like grammar in writing. Say for example we didn’t use periods when we wrote – everyone would have a heck of a time trying to figure out when sentences started and ended. Or if we didn’t employ the comma, how would we convey a dramatic pause to the reader?



And let me tell you, if you ever had an English teacher with an overactive red pen, the compiler is ten times worse. In fact – your programs WILL NOT compile without perfect syntax. This might drive you crazy at first because it is very natural to forget syntax. As you gain experience programming you will learn to be assiduous about coding grammar.



3.1.1 THE SEMICOLON

A semicolon needs to follow every statement written in the Arduino programming language. For example, …

Int LedPin=9;

In this statement, I am assigning a value to an integer variable (we will cover this later), notice the semicolon at the end. This tells the compiler that you have finished a chunk of code and are moving on to the next piece. A semicolon is to Arduino code, as a period is to a sentence. It signifies a complete statement.

3.1.2 THE DOUBLE BACKSLASH FOR SINGLE LINE COMMENTS //

Comments are what you use to annotate code. Good code is commented well. Comments are meant to inform you and anyone else who might stumble across your code, what the heck you were thinking when you wrote it. A good comment would be something like this…

Now, in 3 months when I review this program, I know where to stick my LED. Comments will be ignored by the compiler – so you can write whatever you like in them. If you have a lot you need to explain, you can use a multi-line comment, shown below…

```
//This is an example
```

Comments are like the footnotes of code, except far more prevalent and not at the bottom of the page.

3.1.3 THE CURLY BRACES

Curly braces are used to enclose further instructions carried out by a function (we discuss functions next). There is always an opening curly bracket and a closing curly bracket. If you forget to close a curly bracket, the compiler will not like it and throw an error code.

```
Void loop (){

}
```

Remember – no curly brace may go unclosed!

3.1.4 FUNCTION ( )

Functions are pieces of code that are used so often that they are encapsulated in certain keywords so that you can use them more easily. For example, a function could be the following set of instructions…

This set of simple instructions could be encapsulated in a function that we call WashDog. Every time we want to carry out all those instructions we just type WashDog and voila – all the instructions are carried out. In Arduino, there are certain functions that are used so often they have been built into the IDE. When you type them, the name of the function will appear orange. The function pinMode(), for example, is a common function used to designate the mode of an Arduino pin.

What’s the deal with the parentheses following the function pinMode? Many functions require *arguments* to work. An argument is information the function uses when it runs. For our WashDog function, the arguments might be dog name and soap type, or temperature and size of a bucket.

pinMode(13, OUTPUT);

The argument 13 refers to pin 13, and OUTPUT is the mode in which you want the pin to operate. When you enter these arguments the terminology is called passing. You pass the necessary information to the functions. Not all functions require arguments, but opening and closing parentheses will stay regardless though empty.

Notice that the word OUTPUT is blue. There are certain keywords in Arduino that are used frequently and the color blue helps identify them. The IDE turns them blue automatically. Now we won’t get into it here, but you can easily make your own functions in Arduino, and you can even get the IDE to color them for you. We will, however, talk about the two functions used in nearly EVERY Arduino program.

3.1.5 VOID SETUP ( )

The function, setup(), as the name implies, is used to set up the Arduino board. The Arduino executes all the code that is contained between the curly braces of setup() only once. Typical things that happen in setup() are setting the modes of pins, starting You might be wondering what void means before the function setup(). Void means that the function does not return information. Some functions do return values – our DogWash function might return the number of buckets it required to clean the dog. The function analogRead() returns an integer value between 0-1023. If this seems a bit odd now, don’t worry as we will cover every common Arduino function in depth as we continue the course.

Let us review a couple things you should know about setup()…

1. setup() only runs once.

2. setup() needs to be the first function in your Arduino sketch.

3. setup() must have opening and closing curly braces.

3.1.6 VOID LOOP ( )

You have to love the Arduino developers because the function names are so telling. As the name implies, all the code between the curly braces in loop() is repeated over and over again – in a loop. The loop() function is where the body of your program will reside. As with setup(), the function loop() does not return any values, therefore the word void precedes it.

Does it seem odd to you that the code runs in one big loop? This apparent lack of variation is an illusion. Most of your code will have specific conditions laying in wait which will trigger new actions.

If you have a temperature sensor connected to your Arduino for example, then when the temperature gets to a predefined threshold you might have a fan kick on. The looping code is constantly checking the temperature waiting to trigger the fan. So even though the code loops over and over, not every piece of the code will be executed every iteration of the loop.

3.2 INTRODUCTION ARDUINO LIBRARIES

Libraries are a collection of code that makes it easy for you to connect to a sensor, display, module, etc. For example, the built-in LiquidCrystal library makes it easy to talk to character LCD displays. There are hundreds of additional libraries available on the Internet for download. The built-in libraries and some of these additional libraries are listed in the reference. To use the additional libraries, you will need to install them.

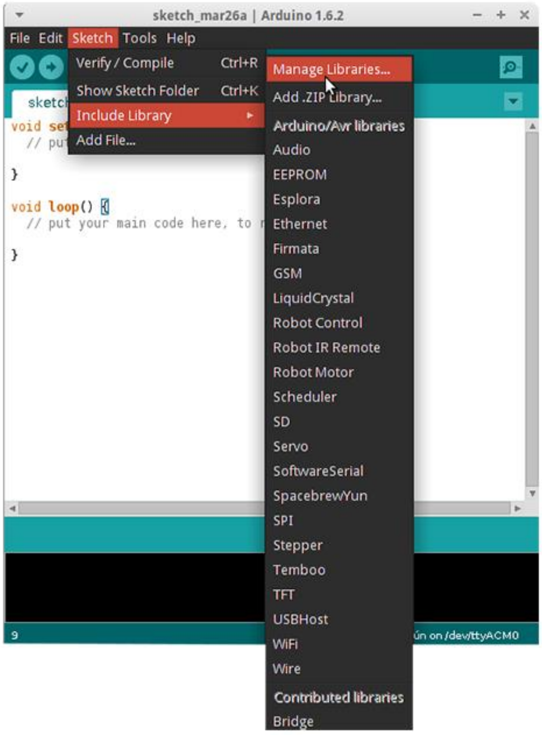
Arduino libraries are managed in three different places: inside the IDE installation folder, inside the core folder and in the libraries folder inside your sketchbook. The way libraries are chosen during compilation is designed to allow the update of libraries present in the distribution. This means that placing a library in the “libraries” folder in your sketchbook overrides the other libraries versions.

The same happens for the libraries present in additional cores installations. It is also important to note that the version of the library you put in your sketchbook may be lower than the one in the distribution or core folders, nevertheless it will be the one used during compilation. When you select a specific core for your board, the libraries present in the core’s folder are used instead of the same libraries present in the IDE distribution folder.

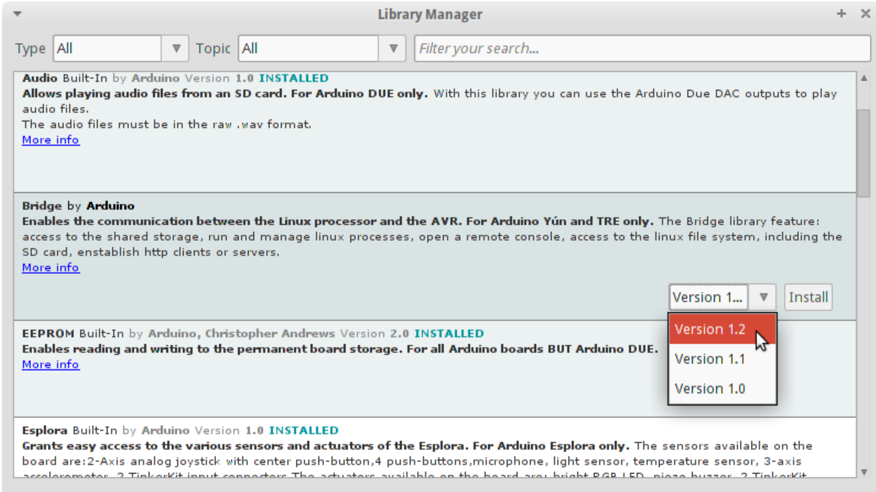
Last, but not least important is the way the Arduino Software (IDE) upgrades itself: all the files in Programs/Arduino (or the folder where you installed the IDE) are deleted and a new folder is created with fresh content. This is why we recommend that you only install libraries to the sketchbook folder so they are not deleted during the Arduino IDE update process.

3.2.1 HOW TO INSTALL A LIBRARY

To install a new library into your Arduino IDE you can use the Library Manager (available from IDE version 1.6.2). Open the IDE and click to the "Sketch" menu and then *Include Library > Manage Libraries*.



Then the Library Manager will open and you will find a list of libraries that are already installed or ready for installation. In this example we will install the Bridge library. Scroll the list to find it, click on it, then select the version of the library you want to install. Sometimes only one version of the library is available. If the version selection menu does not appear, don't worry: it is normal.



Finally click on install and wait for the IDE to install the new library. Downloading may take time depending on your connection speed. Once it has finished, an *Installed* tag should appear next to the Bridge library. You can close the library manager. You can now find the new library available in the *Sketch > Include Library* menu. If you want to add your own library to Library Manager, follow these instructions.

3.3 HOW TO CONNECT ARDUINO BOARD

If you're using a serial board, power the board with an external power supply (6 to 25 volts DC, with the core of the connector positive). Connect the board to a serial port on your computer. On the USB boards, the power source is selected by the jumper between the USB and power plugs. To power the board from the USB port (good for controlling low power devices like LEDs), place the jumper on the two pins closest to the USB plug. To power the board from an external power supply (needed for motors and other high current devices), place the jumper on the two pins closest to the power plug. Either way, connect the board to a USB port on your computer. On Windows, the Add New Hardware wizard will open; tell it you want to specify the location to search for drivers and point to the folder containing the USB drivers you unzipped in the previous step.

The power LED should go on.

3.4 HOW TO UPLOAD A PROGRAM

The content of circuits and Arduino sketches can vary greatly. Before you get started, there is one simple process for uploading a sketch to an Arduino board that you can refer back to.

Follow these steps to upload your sketch:

1. Connect your Arduino using the USB cable.

   The square end of the USB cable connects to your Arduino and the flat end connects to a USB port on your computer.

2. Choose Tools→Board→Arduino Uno to find your board in the Arduino menu.

   You can also find all boards through this menu, such as the Arduino MEGA 2560 and Arduino Leonardo.

3. Choose the correct serial port for your board.

   You find a list of all the available serial ports by choosing Tools→Serial Port→ comX or /dev/tty.usbmodemXXXXX. X marks a sequentially or randomly assigned number. In Windows, if you have just connected your Arduino, the COM port will normally be the highest number, such as com 3 or com 15.

   Many devices can be listed on the COM port list, and if you plug in multiple Arduinos, each one will be assigned a new number. On Mac OS X, the /dev/tty.usbmodem number will be randomly assigned and can vary in length, such as /dev/tty.usbmodem1421 or /dev/tty.usbmodem262471. Unless you have another Arduino connected, it should be the only one visible.

4. Click the Upload button.

   This is the button that points to the right in the Arduino environment. You can also use the keyboard shortcut Ctrl+U for Windows or Cmd+U for Mac OS X.

CHAPTER 4

HARDWARE REQUIREMENTS

Hardware Components of this project are

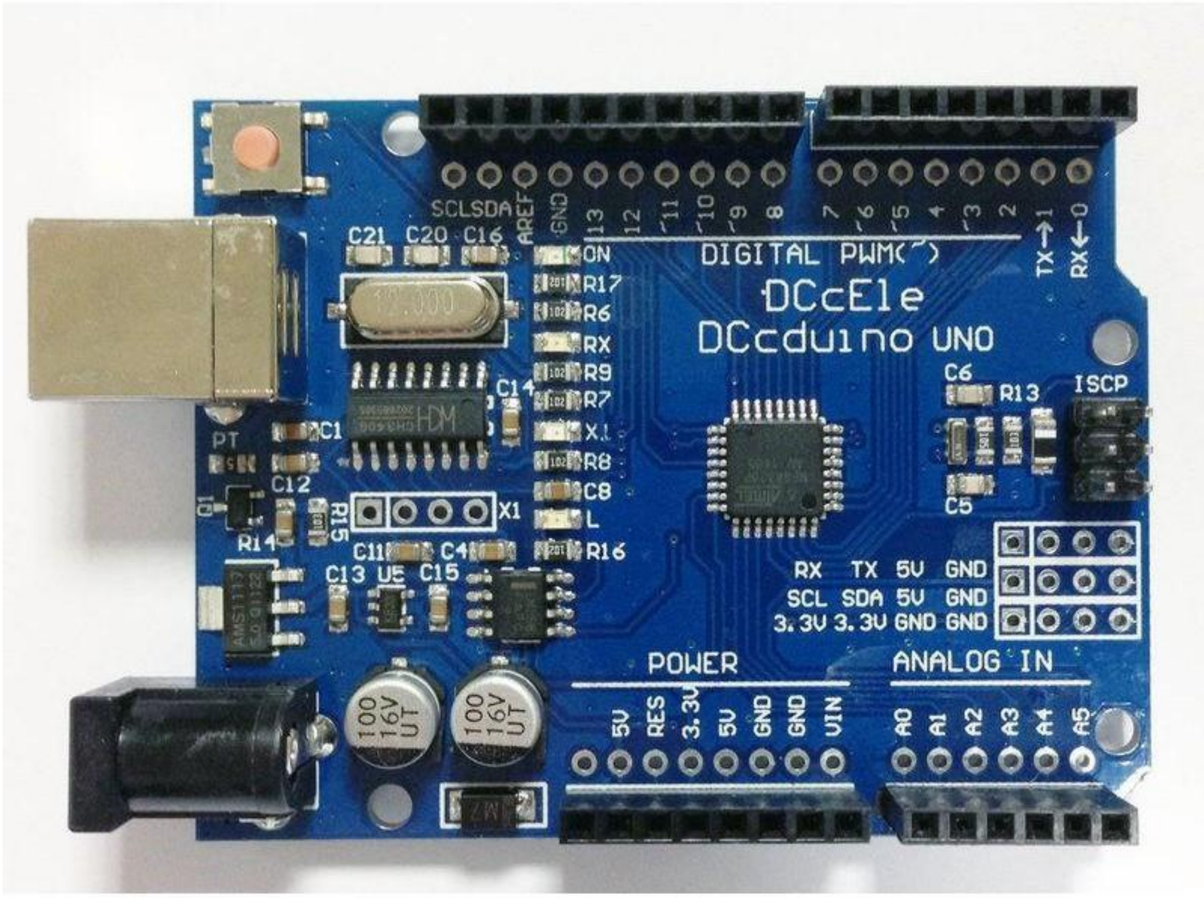
1. ARDIUNO UNO
2. ULTRASONIC SENSOR
3. LCD 20X4
4. L298N
5. WIRELESS CHARGING MODULE
6. LEDS
7. RESISTERS

4.1 INTRODUCTION TO ARDUINO UNO

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing.

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike.

Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without a background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs. The software, too, is open-source, and it is growing through the contributions of users worldwide.



4.1.1 WHY ARDUINO

Thanks to its simple and accessible user experience, Arduino has been used in thousands of different projects and applications. The Arduino software is easy-to-use for beginners, yet flexible enough for advanced users. It runs on Mac, Windows, and Linux. Teachers and students use it to build low cost scientific instruments, to prove chemistry and physics principles, or to get started with programming and robotics. Designers and architects build interactive prototypes, musicians and artists use it for installations and to experiment with new musical instruments. Makers, of course, use it to build many of the projects exhibited at the Maker Faire, for example. Arduino is a key tool to learn new things. Anyone - children, hobbyists, artists, programmers - can start tinkering just following the step by step instructions of a kit, or sharing ideas online with other members of the Arduino community.

4.1.2 ADVANTAGES OF ARDUINO

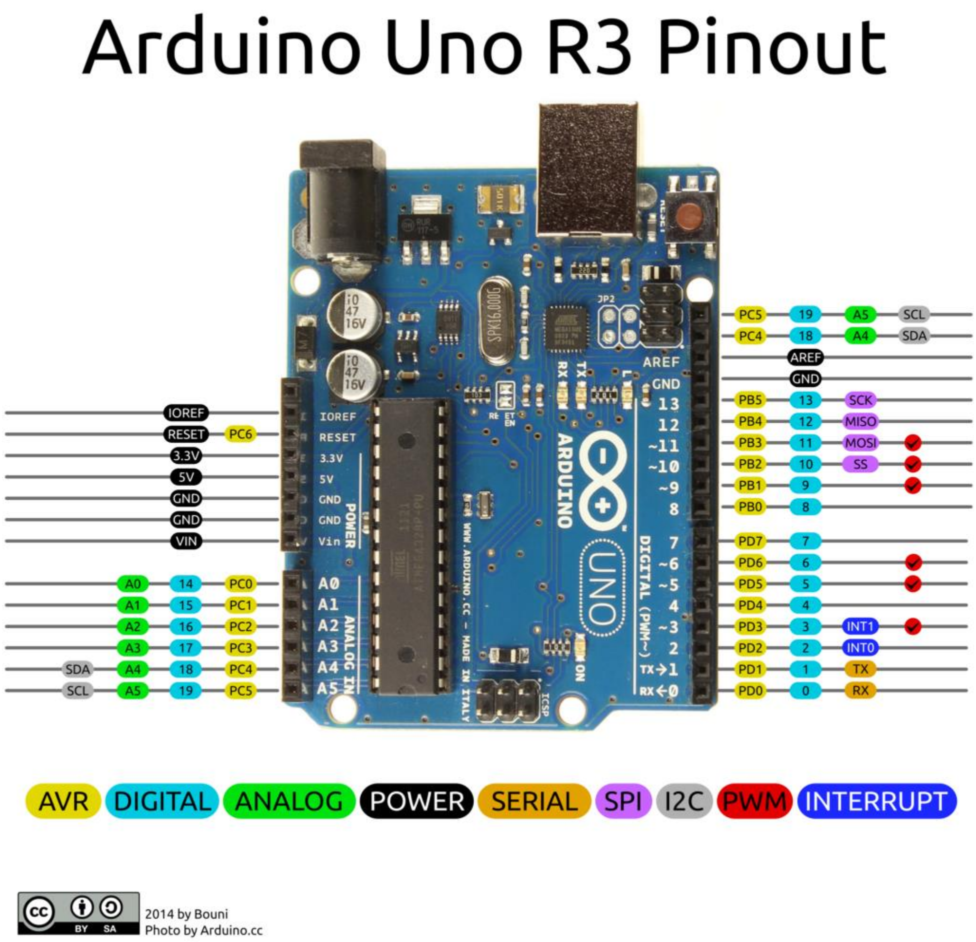
- **Inexpensive** - Arduino boards are relatively inexpensive compared to other microcontroller platforms. The least expensive version of the Arduino module can be assembled by hand, and even the pre-assembled Arduino modules cost less than $50
- **Cross-platform** - The Arduino Software (IDE) runs on Windows, Macintosh OSX, and Linux operating systems. Most microcontroller systems are limited to Windows.
- **Simple, clear programming environment** - The Arduino Software (IDE) is easy-to-use for beginners, yet flexible enough for advanced users to take advantage of as well. For teachers, it's conveniently based on the Processing programming environment, so students learning to program in that environment will be familiar with how the Arduino IDE works.
- **Open source and extensible software** - The Arduino software is published as open source tools, available for extension by experienced programmers. The language can be expanded through C++ libraries, and people wanting to understand the technical details can make the leap from Arduino to the AVR C programming language on which it's based. Similarly, you can add AVR-C code directly into your Arduino programs if you want to.
- **Open source and extensible hardware** - The plans of the Arduino boards are published under a Creative Commons license, so experienced circuit designers can make their own version of the module, extending it and improving it. Even relatively inexperienced users can build the breadboard version of the module in order to understand how it works and save money.

4.1.3 FEATURES OF ARDUINO UNO

The **Arduino Uno** is a microcontroller board based on the ATmega328. Arduino is an open-source, prototyping platform and its simplicity makes it ideal for hobbyists to use as well as professionals. The Arduino Uno has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started.

Features of the Arduino UNO:

- Microcontroller: ATmega328
- Operating Voltage: 5V
- Input Voltage (recommended): 7-12V
- Input Voltage (limits): 6-20V
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- Analog Input Pins: 6
- DC Current per I/O Pin: 40 mA
- DC Current for 3.3V Pin: 50 mA
- Flash Memory: 32 KB of which 0.5 KB used by bootloader
- SRAM: 2 KB (ATmega328)
- EEPROM: 1 KB (ATmega328)
- Clock Speed: 16 MHz



4.2 INTRODUCTION TO ULTRASONIC SENSOR

Ultrasonic transducers or **ultrasonic sensors** are a type of acoustic sensor divided into three broad categories: transmitters, receivers and transceivers. Transmitters convert electrical signals into ultrasound, receivers convert ultrasound into electrical signals, and transceivers can both transmit and receive ultrasound.

In a similar way to radar and sonar, ultrasonic transducers are used in systems which evaluate targets by interpreting the reflected signals. For example, by measuring the time between sending a signal and receiving an echo the distance of an object can be calculated. Passive ultrasonic sensors are basically microphones that detect ultrasonic noise that is present under certain conditions.

4.2.1 APPLICATION AND PERFORMANCE

Ultrasound can be used for measuring wind speed and direction (anemometer), tank or channel fluid level, and speed through air or water. For measuring speed or direction, a device uses multiple detectors and calculates the speed from the relative distances to particulates in the air or water. To measure tank or channel liquid level, and also sea level (tide gauge), the sensor measures the distance (ranging) to the surface of the fluid. Further applications include: humidifiers, sonar, medical ultrasonography, burglar alarms, non-destructive testing and wireless charging.

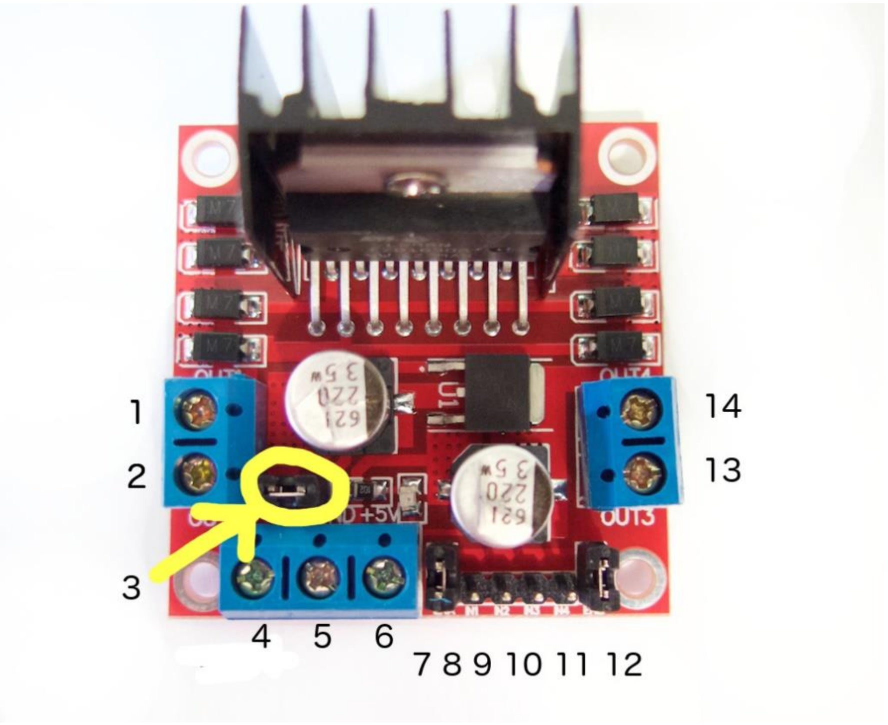
Systems typically use a transducer which generates sound waves in the ultrasonic range, above 18 kHz, by turning electrical energy into sound, then upon receiving the echo turn the sound waves into electrical energy which can be measured and displayed.

The technology is limited by the shapes of surfaces and the density or consistency of the material. Foam, in particular, can distort surface level readings.[1][2]

This technology, as well, can detect approaching objects and track their positions.

4.2 INTRODUCTION TO L293D MOTOR DRIVER

Dual Motor Controller Module 2A with Arduino. This allows you to control the speed and direction of two DC motors, or control one bipolar stepper motor with ease. The L298N H-bridge module can be used with motors that have a voltage of between 5 and 35V DC.
There is also an onboard 5V regulator, so if your supply voltage is up to 12V you can also source 5V from the board. These L298 H-bridge dual motor controller modules .



4.2.1 PIN DESCRIPTION

1. DC motor 1 "+" or stepper motor A+
2. DC motor 1 "-" or stepper motor A-

3. 12V jumper - remove this if using a supply voltage greater than 12V DC. This enables power to the onboard 5V regulator
4. Connect your motor supply voltage here, maximum of 35V DC. Remove 12V jumper if >12V DC
5. GND
6. 5V output if 12V jumper in place, ideal for powering your Arduino (etc)
7. DC motor 1 enable jumper. Leave this in place when using a stepper motor. Connect to PWM output for DC motor speed control.
8. IN1
9. IN2
10. IN3
11. IN4
12. DC motor 2 enable jumper. Leave this in place when using a stepper motor. Connect to PWM output for DC motor speed control.
13. DC motor 2 "+" or stepper motor B+
14. DC motor 2 "-" or stepper motor B-

4.2.2 CONTROLLING DC MOTOR

To control one or two DC motors is quite easy. First connect each motor to the A and B connections on the L298N module. If you're using two motors for a robot (etc) ensure that the polarity of the motors is the same on both inputs. Otherwise you may need to swap them over when you set both motors to forward and one goes backwards!

Next, connect your power supply - the positive to pin 4 on the module and negative/GND to pin 5. If you supply is up to 12V you can leave in the 12V jumper (point 3 in the image above) and 5V will be available from pin 6 on the module. This can be fed to your Arduino's 5V pin to power it from the motors' power supply. Don't forget to connect Arduino GND to pin 5 on the module as well to complete the circuit.

Now you will need six digital output pins on your Arduino, two of which need to be PWM (pulse-width modulation) pins. PWM pins are denoted by the tilde ("~") next to the pin number, for example: Finally, connect the Arduino digital output pins to the driver module. In our example we have two DC motors, so digital pins D9, D8, D7 and D6 will be connected to pins IN1, IN2, IN3 and IN4 respectively. Then connect D10 to module pin 7 (remove the jumper first) and D5 to module pin 12 (again, remove the jumper).

The motor direction is controlled by sending a HIGH or LOW signal to the drive for each motor (or channel). For example for motor one, a HIGH to IN1 and a LOW to IN2 will cause it to turn in one direction, and a LOW and HIGH will cause it to turn in the other direction. However the motors will not turn until a HIGH is set to the enable pin (7 for motor one, 12 for motor two). And they can be turned off with a LOW to the same pin(s). However if you need to control the speed of the motors, the PWM signal from the digital pin connected to the enable pin can take care of it.

Introduction: Wireless Charging Module

The proliferation of electronic devices in our daily lives has created a growing demand for convenient and efficient charging solutions. In response to this need, wireless charging technology has emerged as a game-changing innovation, eliminating the reliance on cables and enabling the seamless charging of devices through the use of wireless charging modules.

A wireless charging module, also known as an inductive charging module, utilizes electromagnetic fields to transfer power between the charging station and compatible devices. It employs the principles of electromagnetic induction, where an alternating current in the charging station generates a magnetic field, which is then picked up by a receiver coil in the device, converting it back into electrical energy for charging.

The introduction of wireless charging modules has revolutionized the way we power our devices. No longer bound by the constraints of physical connectors and cables, wireless charging offers a more convenient and clutter-free charging experience. Users can simply place their compatible devices, such as smartphones, tablets, or wearables, on a wireless charging pad or dock, and the charging process initiates automatically.

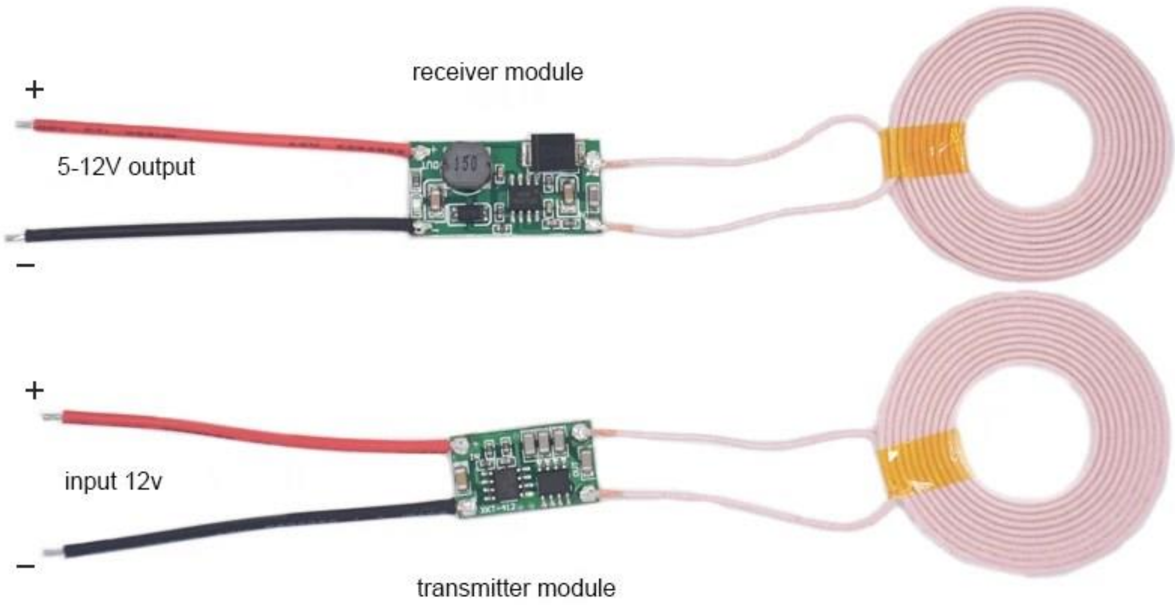
One of the key advantages of wireless charging modules is their versatility and compatibility. Many modern smartphones and other electronic devices come equipped with wireless charging capabilities, enabling seamless integration with wireless charging pads or modules. Additionally, wireless charging modules are designed to support various device types, making them suitable for a wide range of consumer electronics, including smartwatches, headphones, and even electric vehicles.

Wireless charging modules also offer the benefit of safety and durability. With no exposed charging ports or connectors, there is a reduced risk of damage caused by repeated plugging and unplugging of cables. Additionally, wireless charging modules employ safety mechanisms such as foreign object detection, which prevents charging if any non-compatible object is placed on the charging pad, ensuring a safe charging environment.

Another advantage of wireless charging modules is their potential for future advancements. As technology evolves, wireless charging capabilities are expected to improve in terms of charging speed and efficiency. The development of higher power wireless charging modules allows for faster charging times, while ongoing research focuses on extending the range and expanding the charging footprint, enabling charging from a distance without direct contact with the charging pad.

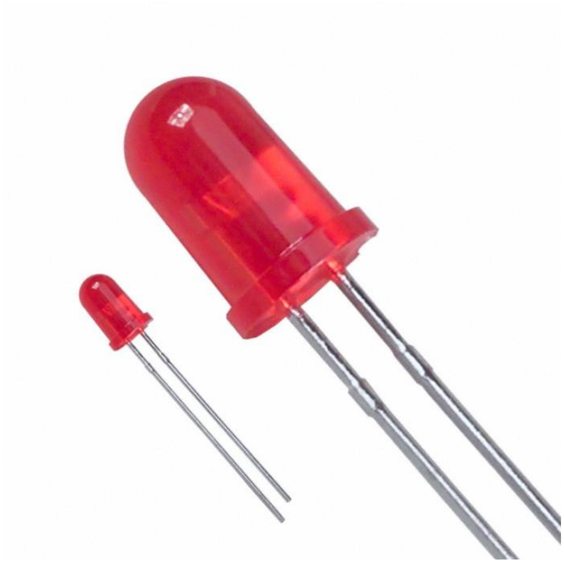
Furthermore, wireless charging modules contribute to the reduction of electronic waste. With traditional charging methods that involve disposable cables and connectors, there is a significant accumulation of electronic waste over time. Wireless charging eliminates the need for disposable cables, promoting a more sustainable and environmentally friendly charging solution.

In conclusion, wireless charging modules have revolutionized the way we charge our electronic devices, providing a convenient, versatile, and clutter-free charging experience. With their compatibility, safety features, and potential for future advancements, wireless charging modules are poised to become the standard charging method for a wide range of consumer electronics. As technology continues to evolve, wireless charging modules hold the promise of faster charging speeds, extended range, and a more sustainable approach to power our devices wirelessly.



4.3 INTRODUCTION TO LEDS

A **light-emitting diode** (**LED**) is a two-lead semiconductor light source. It is a p–n junction diode that emits light when activated. When a suitable current is applied to the leads, electrons are able to recombine with electron holes within the device, releasing energy in the form of photons. This effect is called electroluminescence, and the color of the light (corresponding to the energy of the photon) is determined by the energy band gap of the semiconductor. LEDs are typically small (less than 1 mm^2) and integrated optical components may be used to shape the radiation pattern.



Appearing as practical electronic components in 1962, the earliest LEDs emitted low-intensity infrared light. Infrared LEDs are still frequently used as transmitting elements in remote-control circuits, such as those in remote controls for a wide variety of consumer electronics. The first visible-light LEDs were of low intensity and limited to red. Modern LEDs are available across the visible, ultraviolet, and infrared wavelengths, with very high brightness.

Early LEDs were often used as indicator lamps for electronic devices, replacing small incandescent bulbs. They were soon packaged into numeric readouts in the form of seven-segment displays and were commonly seen in digital clocks. Recent developments have produced LEDs suitable for environmental and task lighting. LEDs have led to new displays and sensors, while their high switching rates are useful in advanced communications technology.

4.3.1 WORKING PRINCIPLE

A P-N junction can convert absorbed light energy into a proportional electric current. The same process is reversed here (i.e. the P-N junction emits light when electrical energy is applied to it). This phenomenon is generally called electroluminescence, which can be defined as the emission of light from a semiconductor under the influence of an electric field. The charge carriers recombine in a forward-biased P-N junction as the electrons cross from the N-region and recombine with the holes existing in the P-region. Free electrons are in the conduction band of energy levels, while holes are in the valence energy band. Thus the energy level of the holes is less than the energy levels of the electrons. Some portion of the energy must be dissipated to recombine the electrons and the holes. This energy is emitted in the form of heat and light.

The electrons dissipate energy in the form of heat for silicon and germanium diodes but in gallium arsenide phosphide (GaAsP) and gallium phosphide (GaP) semiconductors, the electrons dissipate energy by emitting photons. If the semiconductor is translucent, the junction becomes the source of light as it is emitted, thus becoming a light-emitting diode. However, when the junction is reverse biased, the LED produces no light and—if the potential is great enough, the device is damaged.

4.3.2 TECHNOLOGY

The LED consists of a chip of semiconducting material doped with impurities to create a p-n junction. As in other diodes, current flows easily from the p-side, or anode, to the n-side, or cathode, but not in the reverse direction. Charge-carriers—electrons and holes—flow into the junction from electrodes with different voltages. When an electron meets a hole, it falls into a lower energy level and releases energy in the form of a photon.

The wavelength of the light emitted, and thus its color, depends on the band gap energy of the materials forming the p-n junction. In silicon or germanium diodes, the electrons and holes usually recombine by a non-radiative transition, which produces no optical emission, because these are indirect band gap materials. The materials used for the LED have a direct band gap with energies corresponding to near-infrared, visible, or near-ultraviolet light.

LED development began with infrared and red devices made with gallium arsenide. Advances in materials science have enabled making devices with ever-shorter wavelengths, emitting light in a variety of colors.

LEDs are usually built on an n-type substrate, with an electrode attached to the p-type layer deposited on its surface. P-type substrates, while less common, occur as well. Many commercial LEDs, especially GaN/InGaN, also use sapphire substrate.

4.3.3 ADVANTAGES

- **Efficiency:** LEDs emit more lumens per watt than incandescent light bulbs.[150] The efficiency of LED lighting fixtures is not affected by shape and size, unlike fluorescent light bulbs or tubes.

- **Color:** LEDs can emit light of an intended color without using any color filters as traditional lighting methods need. This is more efficient and can lower initial costs.
- **Size:** LEDs can be very small (smaller than 2 mm^2) and are easily attached to printed circuit boards.
- **Warmup time:** LEDs light up very quickly. A typical red indicator LED achieves full brightness in under a microsecond. LEDs used in communications devices can have even faster response times.
- **Cycling:** LEDs are ideal for uses subject to frequent on-off cycling, unlike incandescent and fluorescent lamps that fail faster when cycled often, or high-intensity discharge lamps(HID lamps) that require a long time before restarting.
- **Dimming:** LEDs can very easily be dimmed either by pulse-width modulation or lowering the forward current.[153] This pulse-width modulation is why LED lights, particularly headlights on cars, when viewed on camera or by some people, appear to be flashing or flickering. This is a type of stroboscopic effect.
- **Cool light:** In contrast to most light sources, LEDs radiate very little heat in the form of IR that can cause damage to sensitive objects or fabrics. Wasted energy is dispersed as heat through the base of the LED.
- **Slow failure:** LEDs mostly fail by dimming over time, rather than the abrupt failure of incandescent bulbs.[72]
- **Lifetime:** LEDs can have a relatively long useful life. One report estimates 35,000 to 50,000 hours of useful life, though time to complete failure may be longer.[154] Fluorescent tubes typically are rated at about 10,000 to 15,000 hours, depending partly on the conditions of use, and incandescent light bulbs at 1,000 to 2,000 hours. Several DOEdemonstrations have shown that reduced maintenance costs from this extended lifetime, rather than energy savings, is the primary factor in determining the payback period for an LED product.

4.3.4 DISADVANTAGES

- **Initial price:** LEDs are currently slightly more expensive (price per lumen) on an initial capital cost basis, than other lighting technologies. As of March 2014, at least one manufacturer claims to have reached $1 per kilolumen.[156] The additional expense partially stems from the relatively low lumen output and the drive circuitry and power supplies needed.

- **Temperature dependence:** LED performance largely depends on the ambient temperature of the operating environment – or thermal management properties. Overdriving an LED in high ambient temperatures may result in overheating the LED package, eventually leading to device failure. An adequate heat sink is needed to maintain long life. This is especially important in automotive, medical, and military uses where devices must operate over a wide range of temperatures, which require low failure rates. Toshiba has produced LEDs with an operating temperature range of −40 to 100 °C, which suits the LEDs for both indoor and outdoor use in applications such as lamps, ceiling lighting, street lights, and floodlights.
- **Voltage sensitivity:** LEDs must be supplied with a voltage above their threshold voltage and a current below their rating. Current and lifetime change greatly with a small change in applied voltage. They thus require a current-regulated supply (usually just a series resistor for indicator LEDs).
- **Color rendition:** Most cool-white LEDs have spectra that differ significantly from a black body radiator like the sun or an incandescent light. The spike at 460 nm and dip at 500 nm can cause the color of objects to be perceived differently under cool-white LED illumination than sunlight or incandescent sources, due to metamerism,[158] red surfaces being rendered particularly poorly by typical phosphor-based cool-white LEDs.
- **Area light source:** Single LEDs do not approximate a point source of light giving a spherical light distribution, but rather a lambertian distribution. So LEDs are difficult to apply to uses needing a spherical light field; however, different fields of light can be manipulated by the application of different optics or "lenses". LEDs cannot provide divergence below a few degrees. In contrast, lasers can emit beams with divergences of 0.2 degrees or less

4.3.5 APPLICATIONS

LED uses fall into four major categories:

- Visual signals where light goes more or less directly from the source to the human eye, to convey a message or meaning
- Illumination where light is reflected from objects to give visual response of these objects
- Measuring and interacting with processes involving no human vision

- Narrow band light sensors where LEDs operate in a reverse-bias mode and respond to incident light, instead of emitting light.

4.4 INTRODUCTION TO RESISTORS

A **resistor** is a passive two-terminal electrical component that implements electrical resistance as a circuit element. In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltages, bias active elements, and terminate transmission lines, among other uses. High-power resistors that can dissipate many watts of electrical power as heat, may be used as part of motor controls, in power distribution systems, or as test loads for generators. Fixed resistors have resistances that only change slightly with temperature, time or operating voltage. Variable resistors can be used to adjust circuit elements (such as a volume control or a lamp dimmer), or as sensing devices for heat, light, humidity, force, or chemical activity.

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment. Practical resistors as discrete components can be composed of various compounds and forms. Resistors are also implemented within integrated circuits. The electrical function of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than nine orders of magnitude. The nominal value of the resistance falls within the manufacturing tolerance, indicated on the component.

4.4.1 THEORY OF OPERATION

The behaviour of an ideal resistor is dictated by the relationship specified by Ohm's law:

Ohm's law states that the voltage (V) across a resistor is proportional to the current (I), where the constant of proportionality is the resistance (R). For example, if a 300 ohm resistor is attached across the terminals of a 12-volt battery, then a current of 12 / 300 = 0.04 amperes flows through that resistor.

Practical resistors also have some inductance and capacitance which affect the relation between voltage and current in alternating current circuits.

The ohm (symbol: Ω) is the SI unit of electrical resistance, named after Georg Simon Ohm. An ohm is equivalent to a volt per ampere. Since resistors are specified and manufactured over a very large range of values, the derived units of milliohm (1 mΩ = 10^{-3} Ω), kilohm (1 kΩ = 10^{3} Ω), and megohm (1 MΩ = 10^{6} Ω) are also in common usage.

4.4.2 POWER DISSIPATION

At any instant, the power *P* (watts) consumed by a resistor of resistance *R* (ohms) is calculated

as:

Where *V* (volts) is the voltage across the resistor and *I*(amps) is the current flowing through it. Using Ohm's law, the two other forms can be derived. This power is converted into heat which must be dissipated by the resistor's package before its temperature rises excessively.

Resistors are rated according to their maximum power dissipation. Discrete resistors in solid-state electronic systems are typically rated as 1/10, 1/8, or 1/4 watt. They usually absorb much less than a watt of electrical power and require little attention to their power rating.

Resistors required to dissipate substantial amounts of power, particularly used in power supplies, power conversion circuits, and power amplifiers, are generally referred to as power resistors; this designation is loosely applied to resistors with power ratings of 1 watt or greater. Power resistors are physically larger and may not use the preferred values, color codes, and external packages described below.

If the average power dissipated by a resistor is more than its power rating, damage to the resistor may occur, permanently altering its resistance; this is distinct from the reversible change in resistance due to its temperature coefficient when it warms. Excessive power dissipation may raise the temperature of the resistor to a point where it can burn the circuit board or adjacent components, or even cause a fire. There are flameproof resistors that fail (open circuit) before they overheat dangerously.

Since poor air circulation, high altitude, or high operating temperatures may occur, resistors may be specified with higher rated dissipation than is experienced in service.

All resistors have a maximum voltage rating; this may limit the power dissipation for higher resistance values.

4.4.3 NON IDEAL PROPERTIES

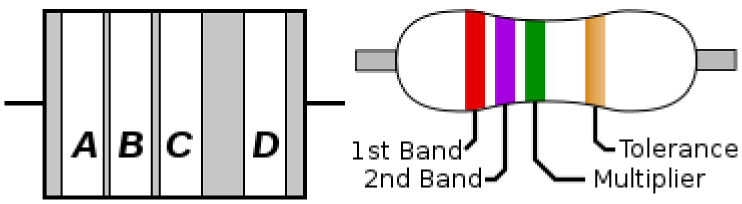
Practical resistors have a series inductance and a small parallel capacitance; these specifications can be important in high-frequency applications. In a low-noise amplifier or pre-amp, the noise characteristics of a resistor may be an issue.

The temperature coefficient of the resistance may also be of concern in some precision applications.

The unwanted inductance, excess noise, and temperature coefficient are mainly dependent on the technology used in manufacturing the resistor. They are not normally specified individually for a particular family of resistors manufactured using a particular technology.[5] A family of discrete resistors is also characterized according to its form factor, that is, the size of the device and the position of its leads (or terminals) which is relevant in the practical manufacturing of circuits using them.

Practical resistors are also specified as having a maximum power rating which must exceed the anticipated power dissipation of that resistor in a particular circuit: this is mainly of concern in power electronics applications. Resistors with higher power ratings are physically larger and may require heat sinks. In a high-voltage circuit, attention must sometimes be paid to the rated maximum working voltage of the resistor. While there is no minimum working voltage for a given resistor, failure to account for a resistor's maximum rating may cause the resistor to incinerate when current is run through it.

4.4.4 RESISTOR COLOUR CODING



To distinguish left from right there is a gap between the C and D bands.

- band **A** is the first significant figure of component value (left side)
- band **B** is the second significant figure (some precision resistors have a third significant figure, and thus five bands).

- band **C** is the decimal multiplier
- band **D** if present, indicates tolerance of value in percent (no band means 20%)

For example, a resistor with bands of yellow, violet, red, and gold has first digit 4 (yellow in table below), second digit 7 (violet), followed by 2 (red) zeros: 4700 ohms. Gold signifies that the tolerance is ±5%, so the real resistance could lie anywhere between 4465 and 4935 ohms.

Resistors manufactured for military use may also include a fifth band which indicates component failure rate (reliability); refer to MIL-HDBK-199[4] for further details.

Tight tolerance resistors may have three bands for significant figures rather than two, or an additional band indicating temperature coefficient, in units of ppm/K.

All coded components have at least two value bands and a multiplier; other bands are optional.

Color	Significant figures	Multiplier	Tolerance		Temp. Coefficient (ppm/K)	
Black	0	$\times 10^0$	–		250	U
Brown	1	$\times 10^1$	±1%	F	100	S
Red	2	$\times 10^2$	±2%	G	50	R
Orange	3	$\times 10^3$	–		15	P
Yellow	4	$\times 10^4$	(±5%)	–	25	Q
Green	5	$\times 10^5$	±0.5%	D	20	Z
Blue	6	$\times 10^6$	±0.25%	C	10	Z
Violet	7	$\times 10^7$	±0.1%	B	5	M
Gray	8	$\times 10^8$	±0.05% (±10%)	A	1	K
White	9	$\times 10^9$	–		–	
Gold	–	$\times 10^{-1}$	±5%	J	–	
Silver	–	$\times 10^{-2}$	±10%	K	–	
None	–	–	±20%	M	–	

LCD 20X4

Certainly! A 20x4 LCD (Liquid Crystal Display) is a type of alphanumeric display commonly used in electronic devices for presenting information. The "20x4" specification refers to the display's size, indicating that it can show 20 characters per line and has 4 lines. Here's a brief explanation of its key features:

1. **Size:** The display consists of 20 characters in each of its 4 lines, providing a total of 80 characters for information display.

2. **Alphanumeric Characters:** The LCD is capable of displaying alphanumeric characters, including letters, numbers, and special symbols.

3**. Backlight:** Many 20x4 LCDs come equipped with a backlight, allowing for visibility in various lighting conditions. The backlight is usually controlled separately from the characters, providing flexibility in terms of contrast and power consumption.

4. **Parallel Interface:** Typically, 20x4 LCDs use a parallel interface to connect with microcontrollers or other electronic devices. This interface allows for the control of each character individually.

5. **HD44780 Controller:** The HD44780 controller is a popular controller used in 20x4 LCDs. It simplifies the interfacing process and allows for easy integration into various projects.

6. **Wide Application:** 20x4 LCDs find applications in a variety of electronic devices, such as microcontroller-based projects, Arduino and Raspberry Pi projects, industrial control panels, measuring instruments, and more.

7. **User-Programmable:** Users can program the content displayed on the LCD based on their specific application requirements. This programmability makes 20x4 LCDs versatile for different projects.

8. **Low Power Consumption:** LCD technology is known for its relatively low power consumption compared to other display technologies, making it suitable for battery-powered devices.

9. **Custom Characters:** Some 20x4 LCDs allow the creation of custom characters, enabling users to design and display characters or symbols beyond the standard set.

In summary, a 20x4 LCD is a versatile and widely used display module known for its ability to present alphanumeric information in a compact form, making it suitable for a broad range of electronic applications.

Certainly! The pinout of a typical 20x4 LCD, especially those using the HD44780 controller, typically includes 16 pins. Here's a brief description of each pin:

1**. VSS (Pin 1):** Ground (0V reference) - Connect this pin to the ground of your circuit.

2. **VDD (Pin 2):** Power Supply Voltage (usually +5V) - Connect this pin to the positive power supply of your circuit.

3**. VO (Pin 3):** Contrast Adjustment - This pin is used to adjust the contrast of the display. Connect it to a variable resistor (potentiometer) for contrast control.

4**. RS (Pin 4):** Register Select - This pin determines whether the data sent to the LCD is a command or actual character data. Connect it to the microcontroller to control this function.

5. **RW (Pin 5):** Read/Write - This pin is used to control the direction of data flow. Connect it to the microcontroller for read or write operations. In many applications, it is tied to ground (write mode) as most of the time you'll be writing data to the display.

6**. E (Pin 6):** Enable - This pin is used to enable the LCD. It triggers the LCD to read the data lines. Connect it to the microcontroller to control this function.

7. **D0 to D7 (Pins 7 to 14):** Data Lines - These pins (D0 to D7) are used for sending 8-bit data to the LCD. In many applications, especially when using 4-bit mode, only D4 to D7 are used.

8**. A (Pin 15):** Backlight Anode - Connect this pin to the positive side of the backlight LED.

9**. K (Pin 16):** Backlight Cathode - Connect this pin to the negative side of the backlight LED.

In many applications, pins VO (contrast), RW (Read/Write), and A and K (backlight) are either connected to fixed values or left unconnected depending on the specific requirements of the project. The RS, E, and data lines (D0 to D7) are typically connected to the microcontroller to facilitate communication with the LCD. Remember to refer to the datasheet of your specific LCD module for accurate pin information as there may be variations based on the manufacturer.


VSS VDD V0 RS RW E D0 D1 D2 D3 D4 D5 D6 D7 A K
Hello,world!
IIC/I2C LCD2004
20 cols, 4 rows
Aitrip

CHAPTER 5

PROGRAM CODE

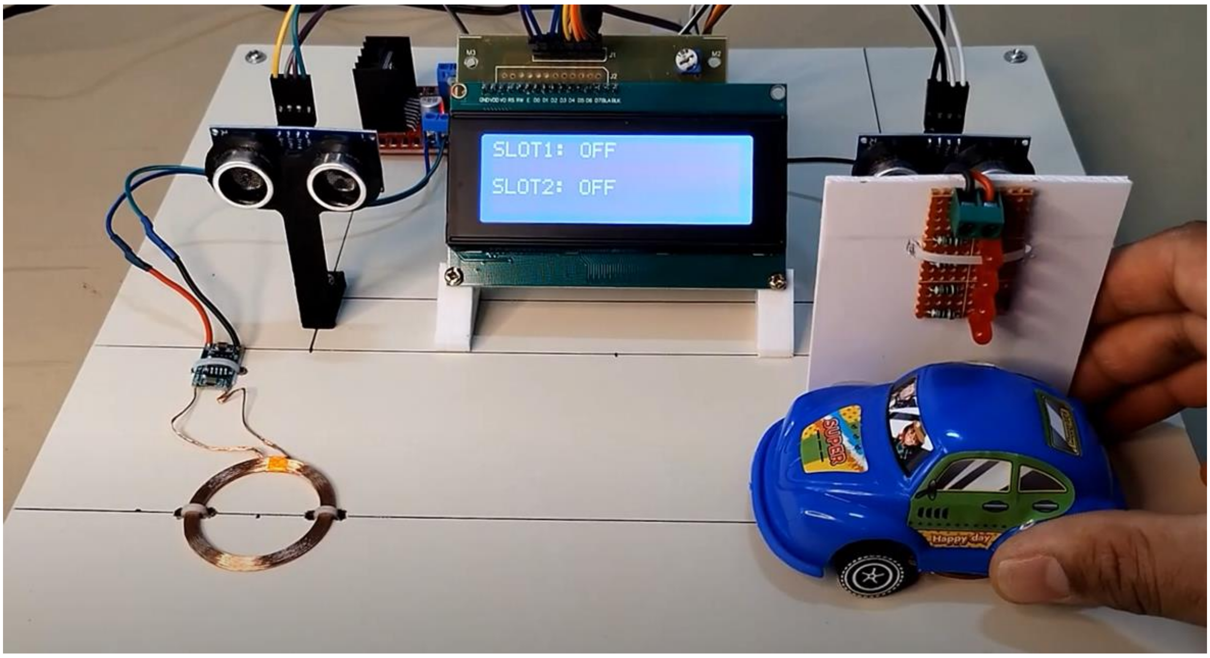
```
#include <Ultrasonic.h>



// Define pins for the Ultrasonic Sensor

#define TRIG_PIN 9

#define ECHO_PIN 10



// Define pins for L298N

#define ENA 5

#define IN1 6

#define IN2 7



// Define variables for distance measurement

Ultrasonic ultrasonic(TRIG_PIN, ECHO_PIN);

long distance;



// Setup function to initialize the pins

void setup() {

  // Initialize serial communication for debugging
```

```
  Serial.begin(9600);

  // Initialize the pins for L298N
  pinMode(ENA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);

  // Initially, stop the L298N motor driver
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 0);
}

// Loop function to continuously check the distance and control the charging
void loop() {
  // Measure the distance to the vehicle
  distance = ultrasonic.Ranging(CM);

  // Print the distance for debugging
  Serial.print("Distance: ");
```

```
  Serial.print(distance);
  Serial.println(" cm");

  // If the vehicle is within 50 cm, start charging
  if (distance <= 50) {
    startCharging();
  } else {
    stopCharging();
  }

  // Wait for a short period before measuring again
  delay(500);
}

// Function to start charging
void startCharging() {
  // Enable the L298N to create the magnetic field in the coil
  digitalWrite(IN1, HIGH);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 255); // Set the speed (PWM) to maximum
```

```
}


// Function to stop charging
void stopCharging() {
  // Disable the L298N to stop the magnetic field in the coil
  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
  analogWrite(ENA, 0);
}
```

CHAPTER 6
RESULT



CHAPTER 7
CONCLUSION

The advent of wireless charging stations for electric vehicles signifies a promising leap forward in the realm of sustainable transportation. With the continuous growth of the electric vehicle market, the need for efficient and user-friendly charging solutions has never been more critical. Wireless charging stations not only provide a convenient and hassle-free method for powering up electric vehicles but also contribute to reducing the overall carbon footprint associated with traditional transportation. As technology advances and becomes more widespread, the integration of wireless charging infrastructure is poised to play a pivotal role in shaping the future of electric mobility. By embracing these cutting-edge technologies, we are taking significant strides towards a cleaner, greener, and more sustainable transportation ecosystem.